Practice workbook · Adding up a finite geometric series

Using S_n = a(1 - r^n)/(1 - r) (including for a sigma sum) to find or show a sum, often in exact or surd form.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-SER-11

1. Question 7 · part c · 4 marks

2019 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part c only.

7 The nth term of a geometric series G is u_n

The first term of G is a and the common ratio of G is r, where $r > 0$

Given that $u_3 = 4$ and that $u_7 = 16$

(a) (i) show that $r = \sqrt{2}$

(ii) find the value of a.

(3)

(b) Find the least value of n for which $u_n > 500$

(4)

The sum of the first n terms of G is S_n

(c) Find S_{20}

Give your answer in the form $p\left(1+\sqrt{2}\right)$ where p is an integer.

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

1. Question 7 · part c · 4 marks

2019 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part c only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 7 · part c · 4 marks

2019 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part c only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 7 · part c · 4 marks

2019 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part c only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 7 is 11 marks)

1. Question 7 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 4 · part i · 4 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part i only.

4 (i) Find the value of $\sum_{k=6}^{17} 3(2)^{k-1}$

(4)

(ii) The common ratio of a geometric series G is positive.

The third term of G is $\frac{7}{13}$ and the ninth term of G is $\frac{448}{9477}$

G is convergent with sum to infinity S

Find the exact value of S

(7)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

2. Question 4 · part i · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 8 · part b · 3 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

8 The nth term of a geometric series G is U_n and the sum of the first n terms of G is S_n

Given that $U_n = \frac{25}{4}\left(\frac{3}{5}\right)^n$

(a) find the exact value of U_5

(1)

(b) Show that $S_n = \sum_{r=1}^{n} \frac{A}{B}\left(\frac{3}{5}\right)^{r-1}$ where A and B are integers to be found.

(3)

The sum to infinity of G is S

(c) Find the least value of n such that $S - S_n < 0.045$

(6)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

3. Question 8 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL